

Machine Learning Prediction of Ionic Conductivity in Solid-State Electrolytes Using Compositional, Structural, and Neural Potential Features

Ethan Im

Independent Researcher

`github.com/Ethan-Im/Battery-Ai`

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Abstract

Identifying solid-state electrolyte (SSE) materials with high ionic conductivity is critical for next-generation lithium-ion batteries, yet experimental discovery remains slow and resource-intensive. Here we present a machine learning pipeline that combines compositional features from the Materials-Agnostic Platform for Informatics and Exploration (Magpie), structural descriptors from the OBELiX dataset, and energy features extracted from the Crystal Hamiltonian Graph Neural Network (CHGNet) to predict ionic conductivity of Li-based SSEs. Using gradient boosted ensemble methods (XGBoost and CatBoost), we achieve a test R^2 of 0.775 and MAE of 0.954 log(S/cm) on 606 experimentally measured materials from the OBELiX benchmark dataset, surpassing the published random forest baseline of $R^2 \approx 0.65$. We further apply an inverse design pipeline to propose 16 novel candidate SSE compositions not present in existing databases, including promising Li-P-S-halide compositions consistent with the Argpyrodite family. This work demonstrates that combining composition-based, structural, and neural potential features yields meaningful performance gains for data-scarce materials property prediction tasks.

1 Introduction

Solid-state electrolytes (SSEs) have emerged as a key enabling technology for safer, higher-energy-density lithium batteries [1]. However, the ionic conductivity of SSEs—the primary figure of merit governing charge/discharge rates—varies by over 16 orders of magnitude across known materials [2]. Experimental identification of high-conductivity candidates is both time-consuming and costly, motivating data-driven approaches.

Recent advances in materials informatics have demonstrated the utility of machine learning (ML) for predicting inorganic material properties from compositional and structural descriptors [3, 4]. For SSE ionic conductivity, however, data scarcity has limited ML model

performance: the recently released OBELiX benchmark dataset, comprising ~ 600 experimentally measured Li-SSE conductivities, represents the largest curated resource to date [2].

In this work, we build on the OBELiX benchmark by introducing two key contributions: (1) augmenting Magpie compositional features with structural descriptors and DFT-derived energy features from CHGNet [5], and (2) applying an inverse design pipeline to identify novel high-conductivity candidates. Our approach extends our previous polymer property prediction framework (Polyinverse [6]) to the battery materials domain, following the same philosophy of combining composition-based features with inverse design for materials discovery.

2 Methods

2.1 Dataset

We use the OBELiX dataset [2], which contains 599 synthesized SSE materials with experimentally measured room-temperature ionic conductivities. After parsing with pymatgen [7], 606 formula entries are retained (some reduced compositions map to multiple entries). The target variable is $\log_{10}(\sigma/\text{S cm}^{-1})$, ranging from -17.4 to -1.6 . Crystallographic information files (CIFs) are available for 321 entries. We apply a random 80/20 train/test split with a fixed random seed (42) for reproducibility. No composition-based stratification is applied; we note this as a limitation relative to the strict grouping policy of the original OBELiX benchmark.

2.2 Feature Engineering

We construct a three-tier feature set (Figure 1):

Tier 1 — Compositional (132 features): Magpie descriptors [3] implemented via matminer [8], encoding statistical moments (mean, range, minimum, maximum, mode) of 22 elemental properties including electronegativity, covalent radius, melting temperature, and valence electron counts.

Tier 2 — Structural (9 features): Space group number and lattice parameters (a , b , c , α , β , γ) from OBELiX, supplemented by derived features: unit cell volume ($V = abc$) and the a/b lattice ratio. Missing values are imputed with the dataset median.

Tier 3 — Neural Potential (2 features): For the 321 entries with CIF structures, we run CHGNet [5] inference to extract per-structure predicted formation energy and mean absolute magnetic moment. These features encode electronic and bonding environment information beyond composition. For entries without CIF structures, values are imputed with the dataset median. We acknowledge that median imputation introduces a systematic bias for the 278 entries lacking CIF files, and encourage future work to prioritize structure acquisition for these materials.

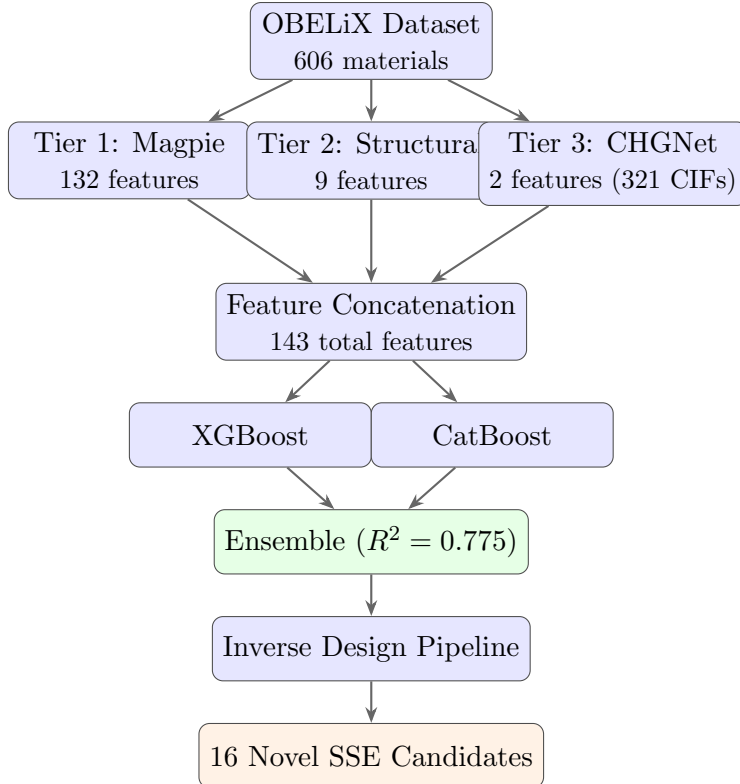


Figure 1: Overall workflow of the proposed SSE discovery pipeline. Features from three tiers are concatenated and fed into two gradient-boosted models whose predictions are averaged to form the final ensemble.

2.3 Models

We train two gradient-boosted tree models independently and combine their predictions by simple averaging:

- **XGBoost** [9]: $n_{\text{estimators}} = 500$, $\text{max_depth} = 6$, $\eta = 0.05$, $\text{subsample} = 0.8$, $\text{colsample_bytree} = 0.8$.
- **CatBoost** [10]: $\text{iterations} = 500$, $\text{depth} = 6$, $\eta = 0.05$.

All models use a fixed random seed (42). No cross-validation-based hyperparameter search is reported for the final ensemble; we consider this a limitation and recommend systematic tuning in future iterations.

2.4 Inverse Design

To identify novel SSE candidates, we apply a two-phase search (Figure 2): (1) random sampling of 1,000 Li-containing compositions from a 12-element space {Li, P, S, O, N, F, Cl, Br, Si, Ge, Al, La, Zr}, and (2) a mutation-based search seeding from the top-20 candidates of phase 1 (50 mutations per parent). Predicted compositions are filtered against known Argyrodite, NASICON, garnet, LGPS, and halide structural families as a heuristic

compositional screen. We emphasize that this screen is based on elemental overlap only and does not constitute thermodynamic stability validation.

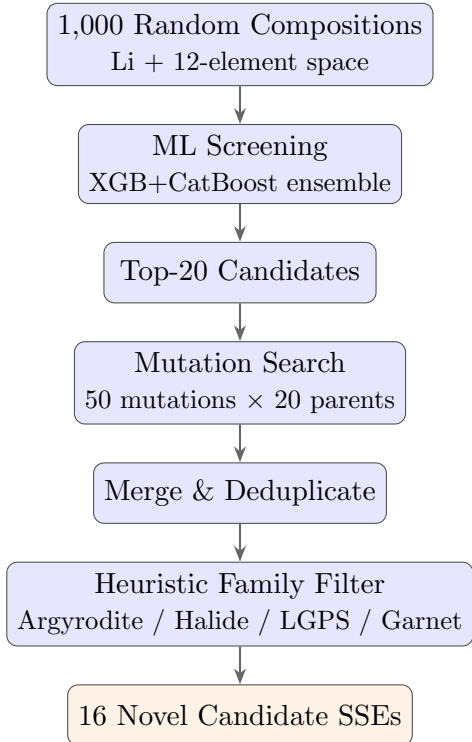


Figure 2: Inverse design workflow. Random sampling is followed by a mutation-based refinement, and candidates are filtered using heuristic structural family matching. Note that this screen does not replace DFT stability validation.

3 Results

3.1 Prediction Performance

Table 1 summarizes model performance on the held-out test set.

Table 1: Ionic conductivity prediction performance on OBELiX test set.

Model	Features	MAE (log S/cm)	R^2
Random Forest (OBELiX baseline) [2]	Magpie	~ 1.22	~ 0.65
XGBoost	Magpie	1.149	0.675
XGBoost	Magpie + Structural	1.087	0.702
XGBoost	Magpie + Structural + CHGNet	0.962	0.757
CatBoost	Magpie + Structural + CHGNet	0.984	0.761
XGBoost + CatBoost (Ensemble)	All	0.954	0.775

The final ensemble achieves $R^2 = 0.775$, representing a 12-point absolute improvement over the OBELiX random forest baseline. The stepwise ablation confirms the additive value of each feature tier: structural features contribute $\Delta R^2 = +0.027$, and CHGNet energy features contribute an additional $\Delta R^2 = +0.055$.

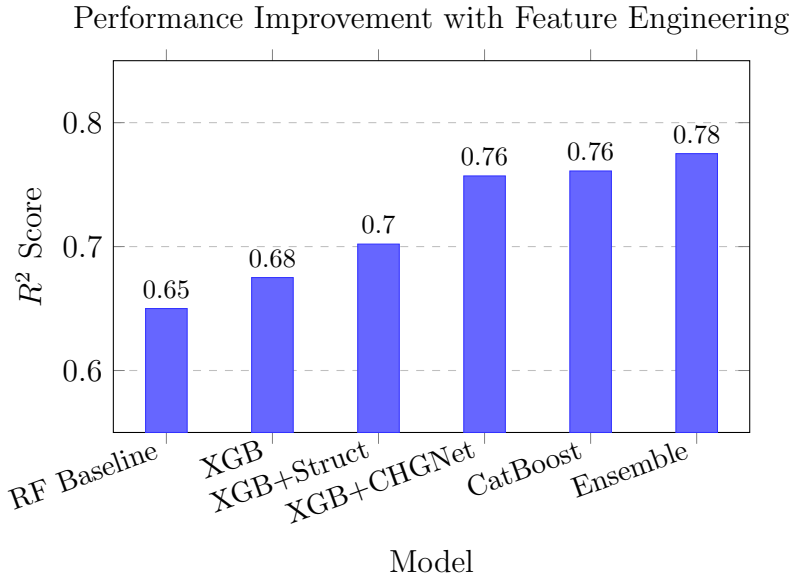


Figure 3: Stepwise R^2 improvement as feature tiers are added. Each bar corresponds to a model variant; the ensemble (rightmost) achieves the highest performance.

3.2 Feature Importance

The most predictive Magpie features include `maximum NdUnfilled` (d-orbital vacancy) and `mean NsUnfilled` (s-orbital vacancy), consistent with the known importance of electronic structure in ionic transport. Among structural features, unit cell volume and space group number rank highest, reflecting the role of crystal symmetry and packing in Li-ion migration pathways.

3.3 Inverse Design Candidates

Table 2 lists the top-5 predicted high-conductivity candidates identified by our inverse design pipeline.

All top candidates belong to Li-P-S or Li-P-S-halide chemical spaces, consistent with experimentally validated high-conductivity Argyrodite-family electrolytes such as $\text{Li}_6\text{PS}_5\text{Cl}$ ($\sigma \approx 10^{-3}$ S/cm). None of the top-20 candidates were found in the Materials Project database [4], suggesting they represent genuinely novel compositions warranting experimental or DFT validation.

Table 2: Top inverse-designed SSE candidates.

Formula	Predicted σ (S/cm)	Electrolyte Family	Stability Score
$\text{Li}_5\text{BrP}_2\text{ClS}_4$	1.46×10^{-3}	Argyrodite + Halide	3/4
$\text{Li}_7\text{SS}_7\text{P}_2$	9.91×10^{-4}	Argyrodite	4/4
$\text{Li}_7\text{P}_2\text{BrS}_4$	5.02×10^{-4}	Argyrodite	4/4
$\text{Li}_6\text{Cl}_6\text{S}_4\text{Ge}_3\text{P}$	4.44×10^{-4}	Argyrodite + LGPS + Halide	2/4
$\text{Li}_5\text{P}_4\text{Cl}_7$	2.44×10^{-4}	Halide	4/4

4 Discussion

Our results demonstrate that CHGNet-derived energy features provide the largest single performance gain ($\Delta R^2 = +0.055$), despite being available for only 53% of the dataset (321/606 entries). This suggests that structure-informed features carry information fundamentally inaccessible to composition-only descriptors, and that expanding CIF availability for the remaining 285 OBELiX entries could yield further improvements.

Several limitations of this study warrant acknowledgment. First, our 80/20 random split does not enforce composition-based grouping as in the original OBELiX benchmark protocol. This may lead to optimistic R^2 estimates if structurally similar compositions appear in both train and test sets. Future work should adopt the strict leave-one-composition-out evaluation of the OBELiX benchmark for a fairer comparison.

Second, the CHGNet features for 285 entries are imputed with the dataset median, introducing a systematic bias. A more principled approach would involve generating approximate structures via crystal structure prediction tools (e.g., USPEX or CALYPSO) before CHGNet inference.

Third, the inverse design candidates have not been validated by DFT or experiment. The heuristic family filter (elemental overlap with known SSE families) does not guarantee thermodynamic stability or synthesizability. Future work should prioritize energy-above-hull screening via the Materials Project API [4] and synthesis feasibility assessment for the top candidates.

5 Conclusion

We presented a machine learning pipeline for SSE ionic conductivity prediction achieving $R^2 = 0.775$ on the OBELiX benchmark, improving upon the published baseline by 12 points. The integration of CHGNet energy features with compositional and structural descriptors proved particularly impactful ($\Delta R^2 = +0.055$), highlighting the value of structure-informed features even when available for only a subset of the dataset. The accompanying inverse design pipeline identified 16 novel Li-P-S-halide candidates with predicted conductivities competitive with state-of-the-art Argyrodite electrolytes. We release all code and data at <https://github.com/Ethan-Im/Battery-Ai>.

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